Heavy Metal Ion Removal with Selective and Fouling-Resistant Polymer Membranes

Sofia Lascano, Nafiseh Khoshnevisan, Ayse Asatekin

Tufts University, Chemical and Biological Engineering Department

Tufts UNIVERSITY | School of Engineering
Chemical and Biological Engineering

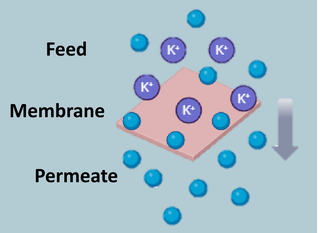
Background and Introduction

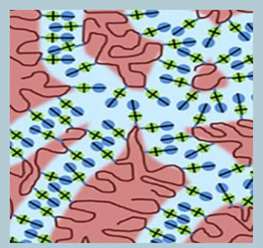
Motivation

- Heavy metals are accumulative, toxic pollutants
- Sources: mining, agricultural (pesticides), e-waste
- Health risks: organ failure, cancer, developmental and reproductive damage
- 15% global energy consumed for traditional chemical separations

Membranes for separations

- Membrane as a low-energy alternative to distillation
- Scalable, environmentally-friendly, cost-effective





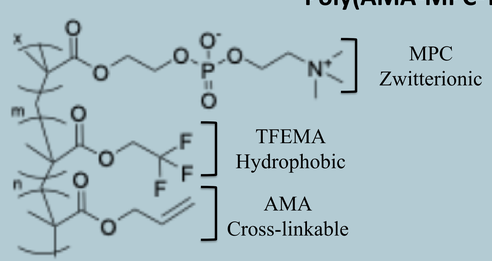
Zwitterions

- Molecule with equal number of negative and positive charges
- Can be polymerized (long, self-assembling chains)
- Form **nanochannels** for ions to pass through

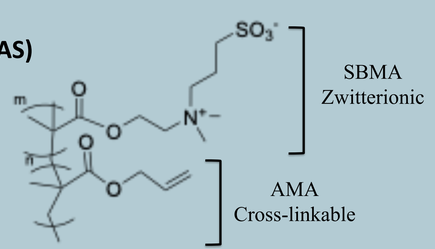
ZI Amphiphilic Copolymer (ZAC) Membranes

- Polymers:

Poly(AMA-MPC-TFEMA) (PAMT)



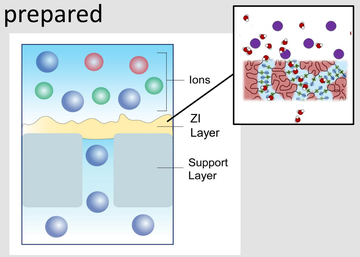
Poly(AMA-SBMA) (PAS)



Methods

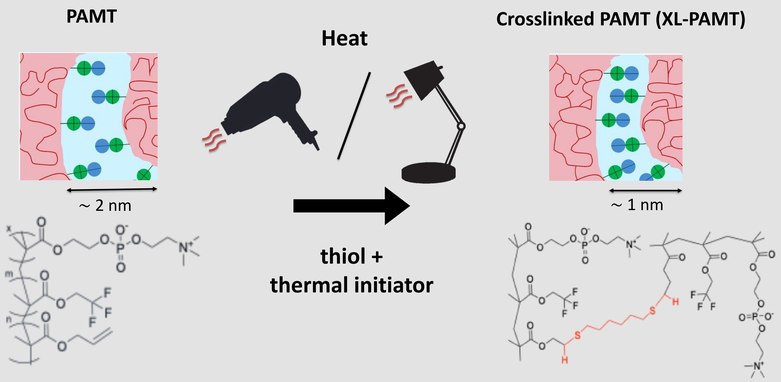
Membrane fabrication

- Thin-film composite (TFC) membranes prepared with PAMT & PAS
- Dissolved in 2,2,2-trifluoroethanol
- Rod-casted onto a porous support (PS35)
- Evaporate solvent using heat gun



Polymer crosslinking

- ZI layer was crosslinked by addition of thiol and thermal initiator
- Both heat lamp and heat gun were tested for comparison

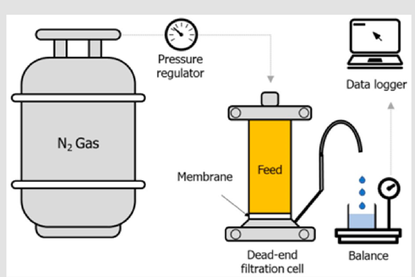


Filtration testing:

- $CuCl_2$, $NiCl_2$, NaCl, $MgCl_2$
- Dead-end filtration cell
- ΔP = 50 psi
- Rejection:

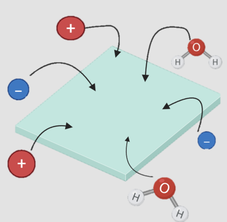
$$R = 100\left(1 - \frac{Cpermeate}{Cfeed}\right)$$

- Characterize pore size with small solute rejection



Membrane characterization

- Polymer film water uptake (swelling)
  → $w_{up} = (m_{wet} - m_{dry})/m_{dry}$
- Salt partitioning coefficient (K_s) compares salt concentration in membrane vs. solution



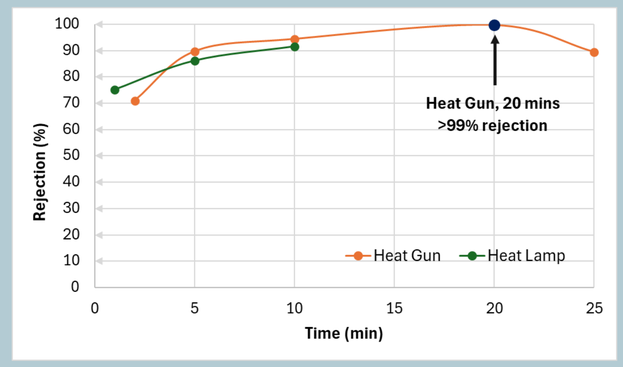
Rejection quantification

- Inductively coupled plasma (ICP) spectroscopy
- Conductivity measurements

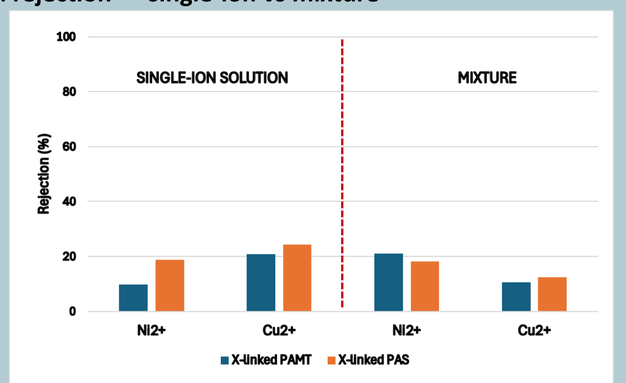
Results

Crosslinking time and membrane performance

- Heat gun > heat lamp for rejection results
  → **20 mins heat gun XL-PAMT, >99% B12 rejection**



Ion rejection — single-ion vs mixture



Water and salt uptake for PAMT films

- Average water uptake: w_{up} = **0.66**
- Average K_s: **0.44**

Conclusion

- Thermal crosslinking using a heat gun can be applied to XL-PAMT and XL-PAS to decrease pore size and improve rejection of single-ions and ion mixtures

Acknowledgements

I would like to thank Prof. Ayse Asatekin, Nafiseh Khoshnevisan, and the rest of the Asatekin Lab Group for their continued guidance and support. I extend my gratitude to Emily Palermo and the entire Summer Scholars team for this great opportunity. Special thanks to the Lee Scholarship for their generous financial support.